

PROJECT REPORT

UNIVERSAL DATA SCIENCE

GUI DASHBOARD

Mini Data Science Project

Submitted By

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Python • NumPy • Pandas • Matplotlib • Seaborn • Tkinter • BeautifulSoup

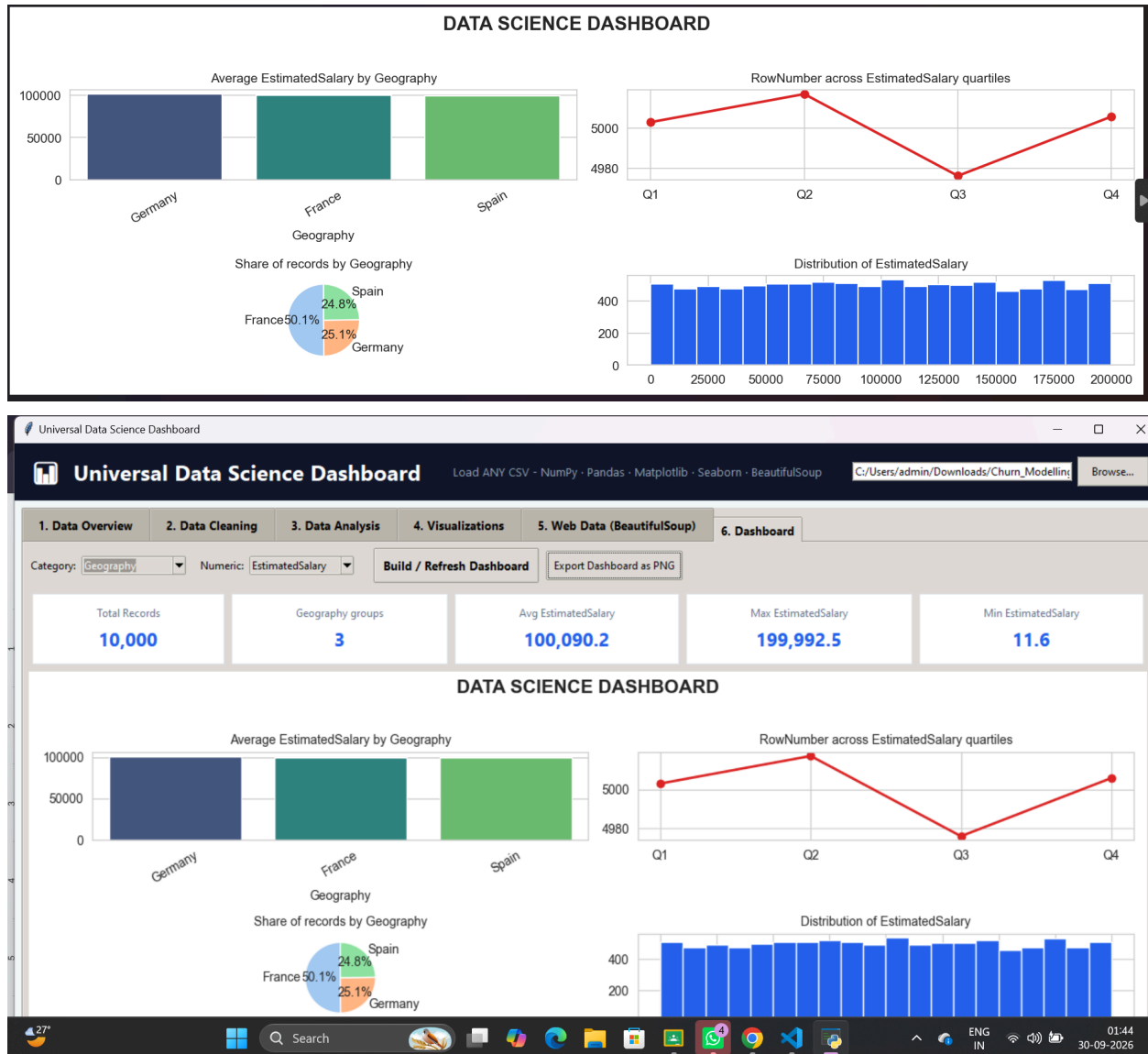
1. Python File

data_science_dashboard.py

2. Dataset

Churn_Modelling.csv / You can use any dataset available.

3. Dashboard



1. Title of the Project

Universal Data Science GUI Dashboard

2. Introduction

The project is a Mini Data Science Project designed as a universal data science GUI dashboard. It is designed to work with different real-world CSV datasets and provides separate tabs for Data Overview, Data Cleaning, Data Analysis, Visualizations, Web Data (BS4), and Dashboard.

The application loads a CSV dataset through the GUI and automatically adapts its analysis and visualizations to the available numeric and categorical columns.

3. Objective

- ❖ To load and explore a CSV dataset.
- ❖ To clean the dataset.
- ❖ To perform data analysis using NumPy and Pandas.
- ❖ To create visualizations using Matplotlib and Seaborn.
- ❖ To provide a dashboard containing KPIs, charts, and automatically generated key findings.
- ❖ To provide a BeautifulSoup-based web-data extraction feature.

4. Dataset Description

The program is designed to load a plain CSV file with a header and at least one numeric column. The code states that the dashboard can be used with datasets such as Sales, E-commerce, Movies, Sports, Student Performance, Employee Data, Healthcare, Weather, Banking, and other suitable real-world datasets.

The program does not define one fixed dataset. It loads the CSV selected by the user or the default dataset.csv when available.

5. Tools and Libraries Used

Library / Tool	Use in the Project
Python	Main programming language.
NumPy	Numerical calculations, statistics and percentile/quartile operations.
Pandas	CSV loading, cleaning, filtering, grouping, sorting and data analysis.
Matplotlib	Charts, figures and dashboard visualizations.
Seaborn	Bar plot, box plot and heatmap visualizations.
Tkinter	Graphical user interface.
BeautifulSoup	Extraction of tabular data from a public web page.
Requests	HTTP request used by the web-data extraction feature.

6. Data Cleaning

The Data Cleaning tab performs the cleaning operations implemented in the program:

- ❖ Strip spaces from column names.
- ❖ Drop exact duplicate rows.
- ❖ Fill missing numeric values with the median.
- ❖ Fill missing text/category values with the mode.
- ❖ Automatically detect and drop ID-like columns.
- ❖ Strip whitespace from text values.
- ❖ Convert suitable low-cardinality text columns to category type.
- ❖ Report numeric and categorical columns after cleaning.

7. Data Analysis

The Data Analysis tab provides a Full Analysis operation containing the following nine operations:

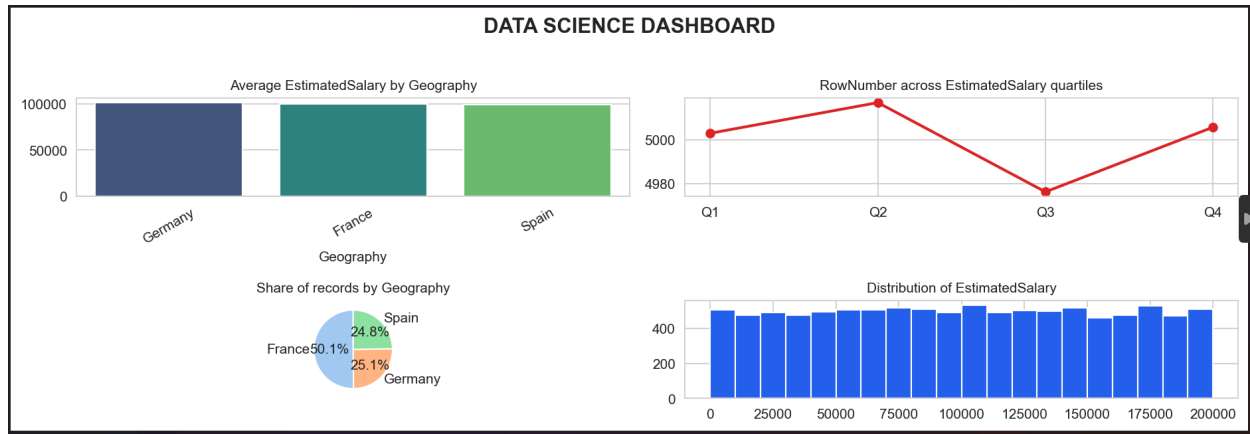
- ❖ Central tendency: mean, median and standard deviation for numeric columns.
- ❖ Range: maximum and minimum values.
- ❖ Sum and count.
- ❖ Value counts for a selected categorical column.
- ❖ Group-wise mean.
- ❖ Group-wise count and sum.
- ❖ Sorting and top-10 results.
- ❖ Correlation matrix.
- ❖ Percentile/quartile bins followed by grouped mean analysis.

8. Data Visualizations

The Visualizations tab provides the chart types implemented in the program.

Chart	Implementation in the Program
Bar Chart	Average numeric value by category.
Line Chart	Numeric trend across quartile bins.
Histogram	Distribution of a numeric column.
Seaborn Box Plot	Numeric values by category.
Pie Chart	Share of records by category.
Seaborn Heatmap	Correlation of numeric columns.

9. Dashboard



10. Key Findings / Observations

The program generates key findings dynamically from the selected dataset. The implemented findings include:

- ❖ Number of records and columns after cleaning.
- ❖ Minimum and maximum values of the selected numeric column.
- ❖ Category with the highest average numeric value.
- ❖ Category with the lowest average numeric value.
- ❖ Most common category and its percentage.
- ❖ Strongest correlation with the selected numeric column when at least two numeric columns are available.
- ❖ Average, median and standard deviation.

11. Conclusion

The Universal Data Science GUI Dashboard combines the data science operations implemented in the program into one graphical application. It provides CSV loading, dataset exploration, data cleaning, statistical analysis, visualizations, dashboard generation, and a bonus web-data extraction feature.

The application is designed to adapt its analysis to the columns available in the selected CSV dataset.